



Numerical investigating the low field NMR response of representative pores at different pulse sequence parameters

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ABSTRACT

We developed a numerical simulation algorithm to explore the nuclear magnetic resonance (NMR) response of the porous media based on the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence and the Bloch equation. The evolution of the magnetization vector of two representative pores at different pulse properties, including the excitation angle, the refocusing angle, the phase angle, as well as the pulse duration are simulated to understand the NMR relaxation signals. The result showed that the normalized magnetization is symmetrical with the excitation angle and positive with the T_2 spectrum's amplitude when the excitation angle is less than 90° . In addition, the refocusing angle has no clear influence on the NMR response. The phase angle of the excitation pulse is inversely correlated with the echo amplitude and can be neglected when the value is lower than 15° . The phase angle of the refocusing pulse causes the zig-zag phenomenon, but the response of the even echoes is not disturbed. Moreover, the influence of the pulse duration should not be neglected at higher values, particularly for the mesopore. The simulation results are helpful for the design and optimization of the pulse sequence, and the data manipulation of the measured signals.

1. Introduction

The low field nuclear magnetic resonance (NMR) technique has been widely applied in geological studies including the pore structure characterization, porosity and permeability prediction, hydrocarbon viscosity estimation, wettability evaluation, as well as the fluid discrimination in recent decades (Freedman et al., 1997; Coates et al., 1999; Dunn et al., 2002; Freedman and Heaton, 2004; Song, 2010; Xie and Xiao, 2011; Xiao et al., 2012; Mitchell et al., 2015; Feng et al., 2016; YanHe et al., 2017; Ge et al., 2018; Liang et al., 2019). The NMR signal is contributed by the hydrogen in pore space, and is lithology-independent in paramagnetic free reservoirs. The most frequently observed parameters both in laboratory and borehole measurements is the transversal relaxation time (T_2), generated by the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence (Coates et al., 1999; Dunn et al., 2002). Despite tremendous successful examples in using the T_2 information for the reservoir evaluation, the measurement and process of low field NMR data to characterize the unconventional resources such as the gas shale and tight sand is still challenging (Bachman et al., 2007;

Grunewald and Knight, 2011; Behroozmand et al., 2015; Washburn, 2014; Mehana and El-monier, 2016; Sondergeld et al., 2016; Obasi and Pashin, 2018; Tan et al., 2019; Testamanti and Rezaee, 2019). Special precautions are necessary before the direct application of the NMR data.

Literature reviews showed that great improvements have been made in the manipulation of the raw decay signals such as the noise reduction (Xie et al., 2014; Ge et al., 2015; Cai and Chen, 2016) and inversion algorithms (Berman et al., 2013; Mitchell and Chandrasekera, 2014; Zou and Xie, 2015; Wang et al., 2017; Guo et al., 2018), and the mathematical transformations of the T_2 spectrum (Ge et al., 2015; Hu et al., 2016; Anand, 2017; Jiang et al., 2019). However, the forward modeling works to investigate the influence of the pulse sequence property are relative rare (Hurlimann and Griffin, 2000; Borneman et al., 2010; Li et al., 2013; Casabianca et al., 2014). Pulse imperfections of the CPMG echo trains may introduce large artifacts (Mandal et al., 2015; Konuma et al., 2018), whereas the intrinsic mechanism still remains ambiguous. This work focuses the influences of the pulse sequence on the NMR measurements and their inversion results for reservoir rocks with different pore size distributions through numerical simulations. We are

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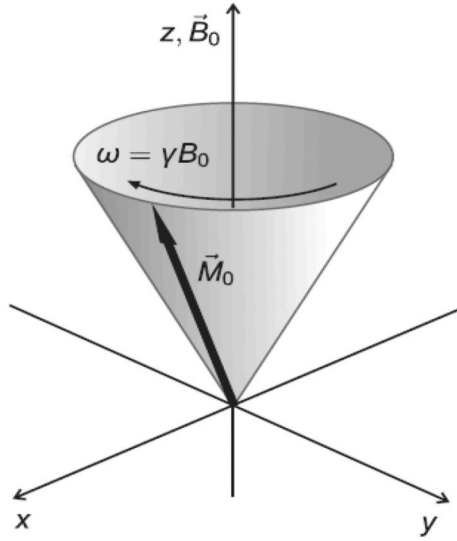


Fig. 1. Larmor precession of the proton about the main magnetic field.

aiming to address the relationship between the NMR response and the properties of the CPMG pulse sequence, and to optimize and compensate the imperfection of the pulse sequence during the acquisition. The rest of paper is organized as follows. In section 2, we review the NMR relaxation briefly. In section 3, we conduct the numerical simulation to investigate the NMR responses at different pulse sequences. We illustrate the conclusion and discussion in section 4. The Butler-Reed-Dawson (BRD) algorithm (Butler et al., 1981; Su et al., 2019) is used to invert the NMR decay curve to the T_2 spectrum.

2. Theoretical background

Detailed interpretations of the NMR relaxation theory and data processing methods are not elaborated here. We just give some basic information of the Bloch equation, the CPMG pulse sequence, and the matrix representation. For simplicity, the effect of the field inhomogeneity and the molecular diffusion are not discussed here.

2.1. The Bloch equation

Neglecting the diffusion term, the vector representation of the well-known Bloch equation that is used to describe the magnetic resonance phenomenon in Cartesian coordinates is expressed as (Bittoun et al., 1984; Haacke et al., 1999; Bhalekar et al., 2011; Walbrecker et al., 2011; Jurczuk et al., 2013; Davies et al., 2014; Behroozmand et al., 2015; Singh, 2016),

$$\frac{d\mathbf{M}}{dt} = \gamma \mathbf{M} \times \mathbf{B} - \frac{M_x \mathbf{x} + M_y \mathbf{y}}{T_2} - \frac{(M_z - M_0) \mathbf{z}}{T_1} \quad (1)$$

where M_x, M_y, M_z are the components of the magnetic moment $\mathbf{M}$ at the time t , T_1 and T_2 are the longitudinal (spin-lattice) and the transversal (spin-spin) relaxation times, and M_0 is the magnetic moment at thermal equilibrium in the main magnetic field B_0 aligned with the z axis, and $\mathbf{B}$ is the alternating magnetic field at the time t , γ is the gyromagnetic ratio of the proton. For the hydrogen, the value is 42.58 MHz/T.

As is shown in Fig. 1, the spins will rotate about the main magnetic field B_0 with the Larmor frequency during the thermal equilibrium state can be expressed as,

$$\omega_0 = 2\pi f_0 = \gamma B_0 \quad (2)$$

where the units of ω_0 and f_0 are rad/s and Hz, respectively.

A carefully constructed radio frequency (RF) field $\mathbf{B}_1$ at a very characteristic pulsation determined by the Larmor frequency is needed to excite and manipulate the spins away from the equilibrium state (Balac and Chupin, 2008). When the RF field is switched on, an interaction occurs between $\mathbf{B}_1$ and $\mathbf{M}$ in the rotating frame, known as the resonance process. The position of $\mathbf{M}$ at the end of the resonance process is determined by the duration of the RF field. Typically this duration is chosen so that the flip angle between the initial position and the resulting one is 90° or 180° . When the RF field is switched off, $\mathbf{M}$ tends to return to its equilibrium position in a process called the relaxation. The flip angle can be expressed as,

$$\theta = \gamma |\mathbf{B}_1| t_\theta = \omega_1 t_\theta = 2\pi f_1 t_\theta \quad (3)$$

where t_θ is the duration of the RF, and $|\mathbf{B}_1|$ is the magnitude of the RF field, the units of ω_1 and f_1 are rad/s and Hz, respectively.

2.2. Solution by the rotation matrix

Essentially, most of RF fields are applied in either x axis or y axis, the resonance and the precession processes can be characterized by the vector rotation. As is seen in Fig. 2, when a RF field with the flip angle of 90° is applied on the y axis, the magnetic moment is then flipped into the x axis, followed by the left-hand rule.

Generally, the rotation matrix for a vector along with x , y , and z axis at a given angle θ can be expressed as,

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \quad (4)$$

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \quad (5)$$

$$R_z(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (6)$$

Moreover, the rotation matrix for a vector along with an arbitrary axis $\mathbf{n}$ (n_x, n_y, n_z) at a given angle θ can be expressed as,

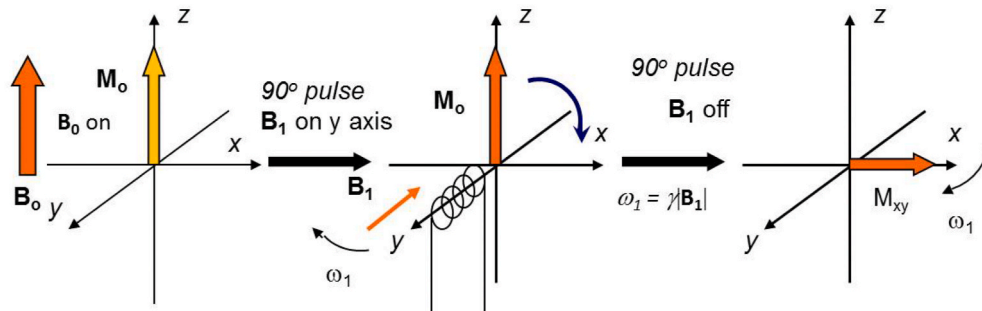


Fig. 2. Flip and rotation of the magnetic vector under the RF field.

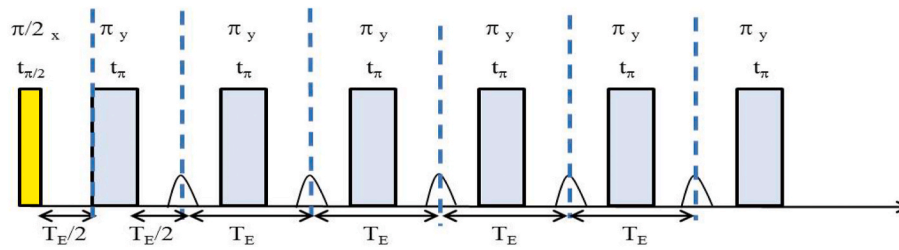
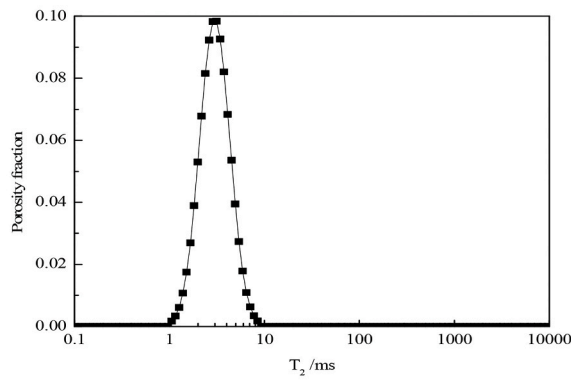
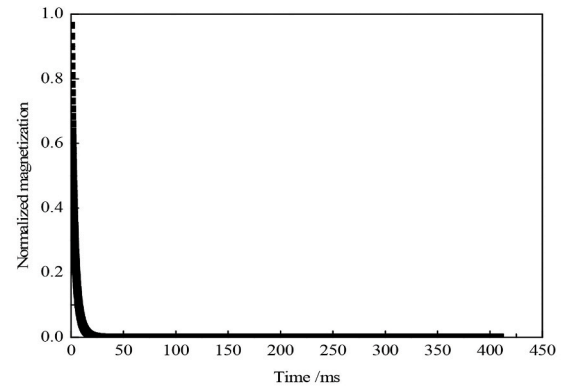
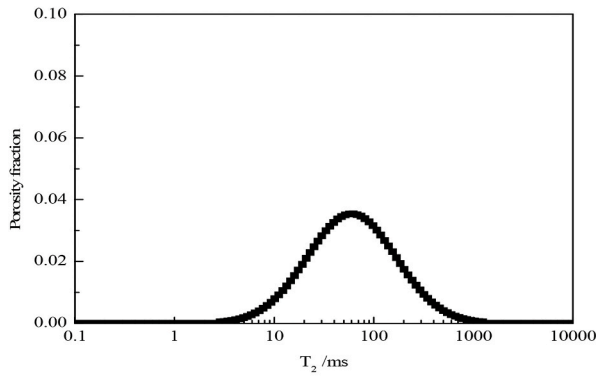
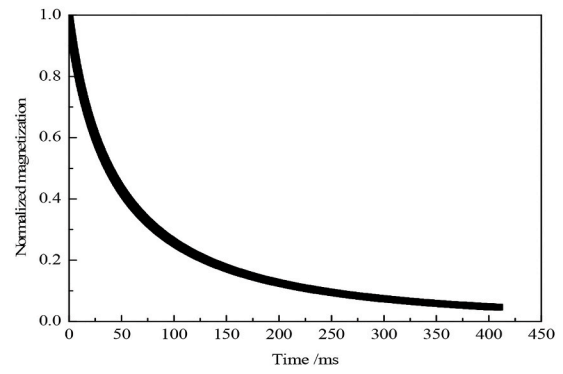


Fig. 3. Diagram of the standard CPMG pulse sequence.

(a) NMR T_2 spectrum of the micropore

(b) NMR decay curve of the micropore

(c) NMR T_2 spectrum of the mesopore

(d) NMR decay curve of the mesopore

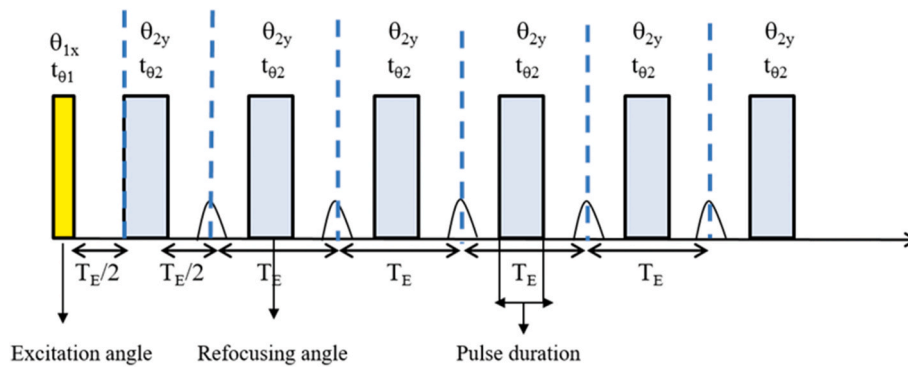
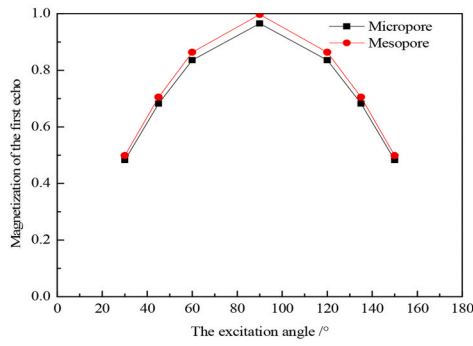
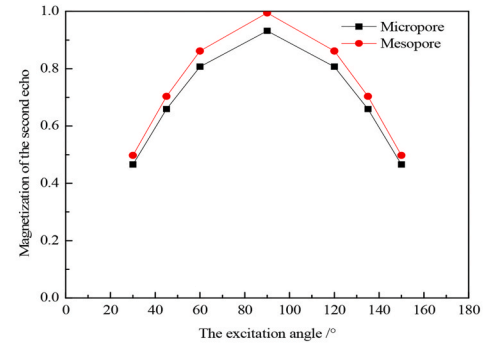
Fig. 4. T_2 spectrums and corresponding decay curves of two forward models.

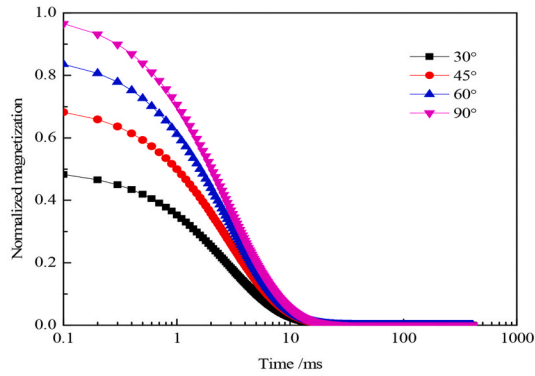
Fig. 5. Diagram of the simulating parameters in the CPMG-like pulse sequence.



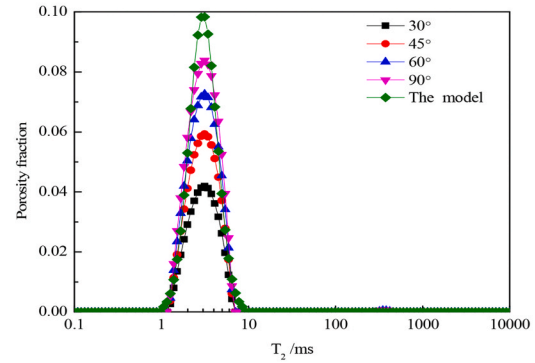
(a) Relationship between the amplitude of the first echo and the excitation angle



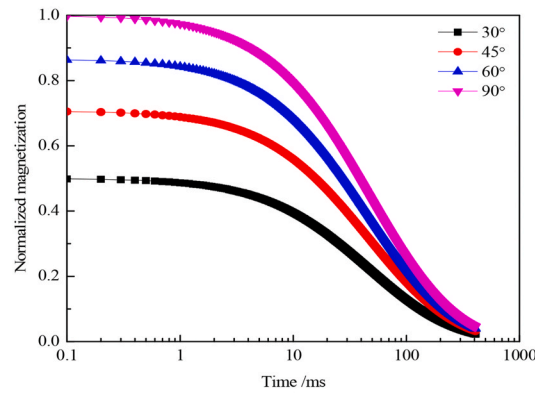
(b) Relationship between the amplitude of the second echo and the excitation angle



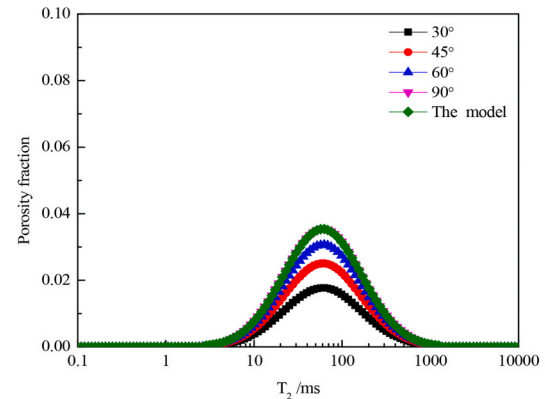
(c) NMR decay curves of the micropore at different excitation angles



(d) T₂ spectrums of the micropore at different excitation angles

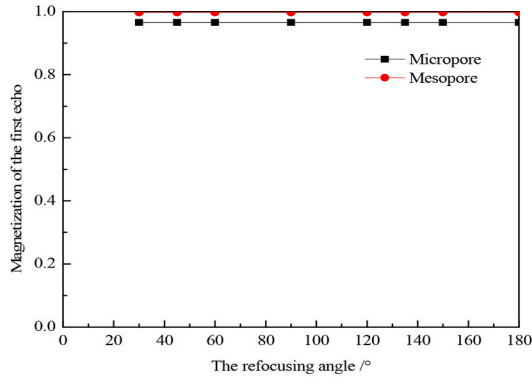


(e) NMR decay curves of the mesopore at different excitation angles

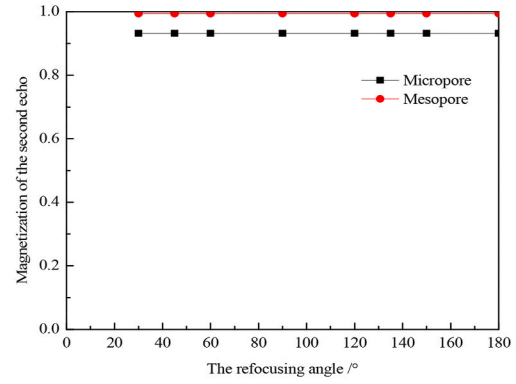


(f) T₂ spectrums of the mesopore at different excitation angles

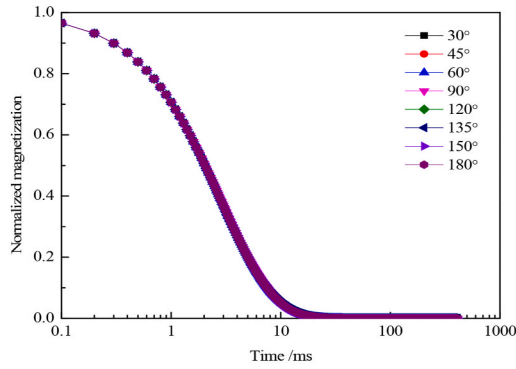
Fig. 6. Decay curves and inversion results at different excitation angles.



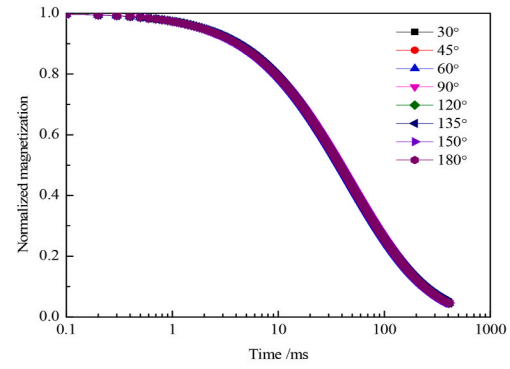
(a) Relationship between the amplitude of the first echo and the refocusing angle



(b) Relationship between the amplitude of the second echo and the refocusing angle



(c) NMR decay curves of the micropore at different refocusing angles



(d) NMR decay curves of the mesopore at different refocusing angles

Fig. 7. Decay curves at different refocusing angles.

$$R_n(\theta) = \begin{vmatrix} n_x^2(1 - \cos \theta) + \cos \theta & n_x n_y(1 - \cos \theta) + n_z \sin \theta & n_x n_z(1 - \cos \theta) - n_y \sin \theta \\ n_x n_y(1 - \cos \theta) - n_z \sin \theta & n_y^2(1 - \cos \theta) + \cos \theta & n_y n_z(1 - \cos \theta) + n_x \sin \theta \\ n_x n_z(1 - \cos \theta) + n_y \sin \theta & n_y n_z(1 - \cos \theta) - n_x \sin \theta & n_z^2(1 - \cos \theta) + \cos \theta \end{vmatrix} \quad (7)$$

According to the rotation matrices mentioned above, the magnetization vector evolution can be computed through the following equation iteratively (Andrade et al., 2011; Gai et al., 2013; Marty et al., 2015; Kose and Kose, 2017),

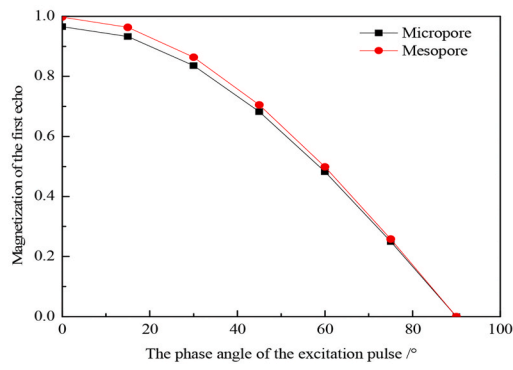
$$\begin{vmatrix} M_x(t + \Delta t) \\ M_y(t + \Delta t) \\ M_z(t + \Delta t) \end{vmatrix} = \begin{vmatrix} e^{-\Delta t/T_2} & 0 & 0 \\ 0 & e^{-\Delta t/T_2} & 0 \\ 0 & 0 & e^{-\Delta t/T_1} \end{vmatrix} \cdot R_{RF}(\theta, \varphi) \cdot \begin{vmatrix} M_x(t) \\ M_y(t) \\ M_z(t) \end{vmatrix} + \begin{vmatrix} 0 \\ 0 \\ 1 - e^{-\Delta t/T_1} \end{vmatrix} \cdot M_0 \quad (8)$$

where $R_{RF}(\theta, \varphi)$ represents the rotating effect of an RF pulse with the flip angle of θ and the phase angle of φ . It can be expressed as,

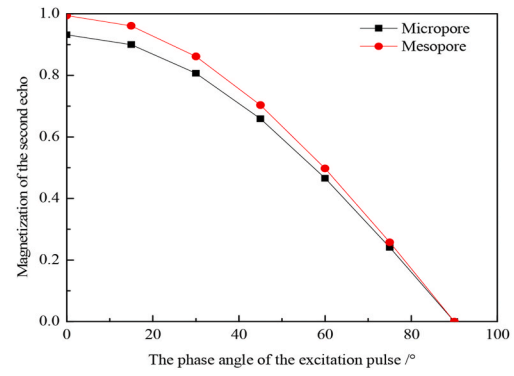
$$R_{RF}(\theta, \varphi) = R_z(-\varphi) \cdot R_x(\theta) \cdot R_z(\varphi) = \begin{vmatrix} \cos^2 \varphi + \sin^2 \varphi \cos \theta & \cos \varphi \sin \varphi - \cos \varphi \sin \varphi \cos \theta & -\sin \varphi \sin \theta \\ \cos \varphi \sin \varphi - \cos \varphi \sin \varphi \cos \theta & \sin^2 \varphi + \cos^2 \varphi \cos \theta & \cos \varphi \sin \theta \\ \sin \varphi \sin \theta & -\cos \varphi \sin \theta & \cos \theta \end{vmatrix} \quad (9)$$

It degenerates to $R_x(\theta)$ when $\varphi = 0$ and $R_y(\theta)$ when $\varphi = \pi/2$.

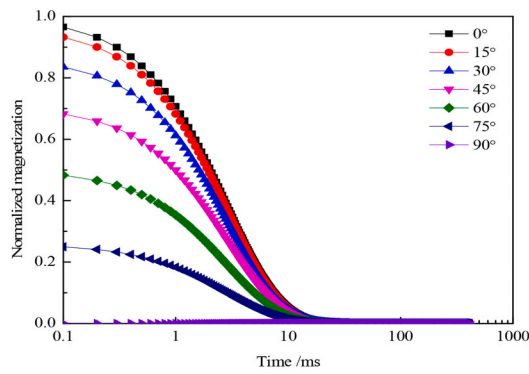
The standard CPMG pulse sequence consists of an initial excitation $\pi/2$ pulse followed by a series of π refocusing pulses (Koroleva et al., 2013; Mandal et al., 2015). The phase of the excitation pulse and the phase of the refocusing pulse are orthogonal, as is shown in Fig. 3. The time between two adjacent refocusing pulses is defined as the echo



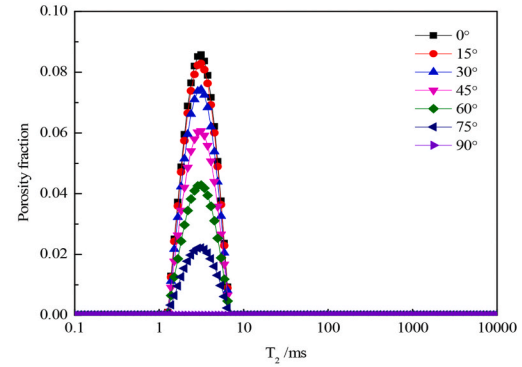
(a) Relationship between the amplitude of the first echo and the phase angle



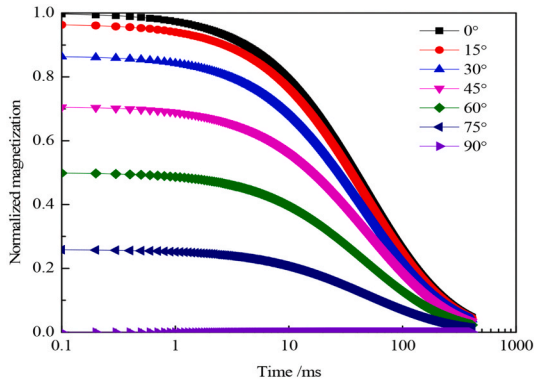
(b) Relationship between the amplitude of the second echo and the phase angle



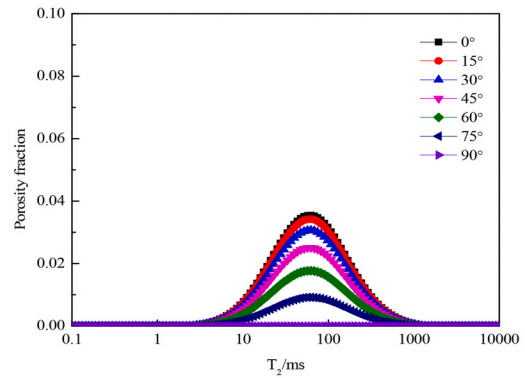
(c) NMR decay curves of the micropore at different phase angles



(d) T₂ spectrums of the micropore at different phase angles



(e) NMR decay curves of the mesopore at different phase angles

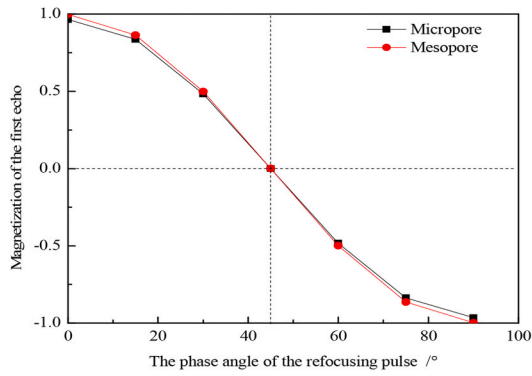


(f) T₂ spectrums of the mesopore at different phase angles

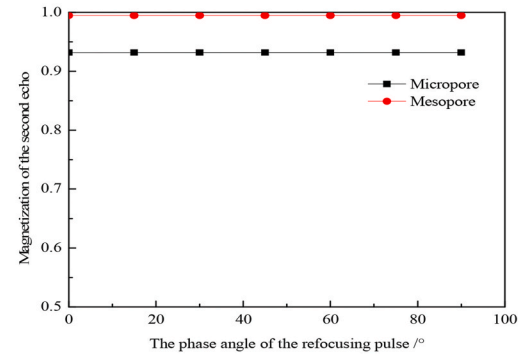
Fig. 8. Decay curves and inversion results of different excitation pulse's phase angles.

spacing (T_E), which is the same as the interval between two adjacent echoes. The duration of the excitation pulse and the refocusing pulse are t_{90} and t_{180} , which can be predicted using equation (3). It is noted that T_E

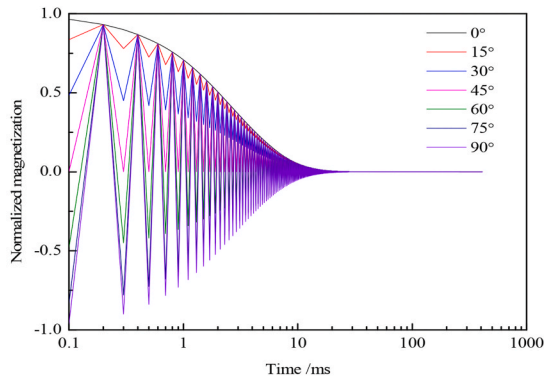
should be much shorter than the transverse relaxation time. Considering the effect of the pulse duration, the magnetization before the first refocusing pulse is expressed as,



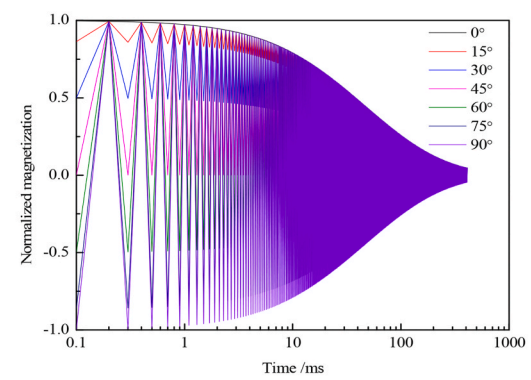
(a) Relationship between the amplitude of the first echo and the phase angle



(b) Relationship between the amplitude of the second echo and the phase angle



(c) NMR decay curves of the micropore at different phase angles



(d) NMR decay curves of the mesopore at different phase angles

Fig. 9. Decay curves of different refocusing pulse's phase angles.

$$\begin{pmatrix} M_x(0) \\ M_y(0) \\ M_z(0) \end{pmatrix} = \begin{pmatrix} e^{-(T_E/2+t_{\theta_1})/T_2} & 0 & 0 \\ 0 & e^{-(T_E/2+t_{\theta_1})/T_2} & 0 \\ 0 & 0 & e^{-(T_E/2+t_{\theta_1})/T_1} \end{pmatrix} \cdot R_{RF}(\theta_1, \varphi) \cdot M_0 + \begin{pmatrix} 0 \\ 0 \\ 1 - e^{-(T_E/2+t_{\theta_1})/T_1} \end{pmatrix} \cdot M_0 \quad (10)$$

$$\begin{pmatrix} M_x(1) \\ M_y(1) \\ M_z(1) \end{pmatrix} = \begin{pmatrix} e^{-(T_E/2+t_{\theta_2})/T_2} & 0 & 0 \\ 0 & e^{-(T_E/2+t_{\theta_2})/T_2} & 0 \\ 0 & 0 & e^{-(T_E/2+t_{\theta_2})/T_1} \end{pmatrix} \cdot R_{RF}(\theta_2, \varphi) \cdot \begin{pmatrix} M_x(0) \\ M_y(0) \\ M_z(0) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 - e^{-(T_E/2+t_{\theta_2})/T_1} \end{pmatrix} \cdot M_0 \quad (11)$$

$$\begin{pmatrix} M_x(k) \\ M_y(k) \\ M_z(k) \end{pmatrix} = \begin{pmatrix} e^{-(T_E/2+t_{\theta_2})/T_2} & 0 & 0 \\ 0 & e^{-(T_E/2+t_{\theta_2})/T_2} & 0 \\ 0 & 0 & e^{-(T_E/2+t_{\theta_2})/T_1} \end{pmatrix} \cdot R_{RF}(\theta_2, \varphi) \cdot \begin{pmatrix} M_x(k-1) \\ M_y(k-1) \\ M_z(k-1) \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 - e^{-(T_E/2+t_{\theta_2})/T_1} \end{pmatrix} \cdot M_0 \quad (12)$$

The magnetization of the first and the k -th ($k \geq 2$) echo are given by, where θ_1 and θ_2 are the excitation angle and the refocusing angle, respectively; t_{θ_1} and t_{θ_2} are their corresponding pulse durations.

For some circumstances, the field inhomogeneity of the static main magnetic field and the radio frequency field should be considered. There are possible two main causes of the field inhomogeneity, including the

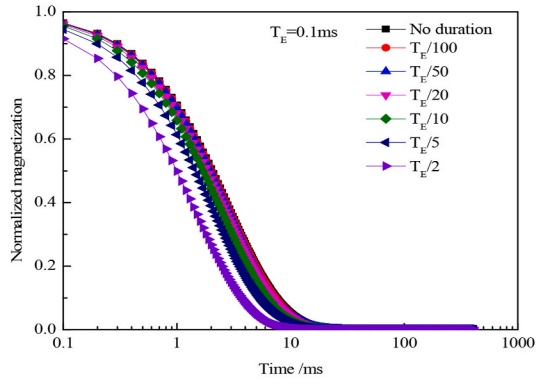
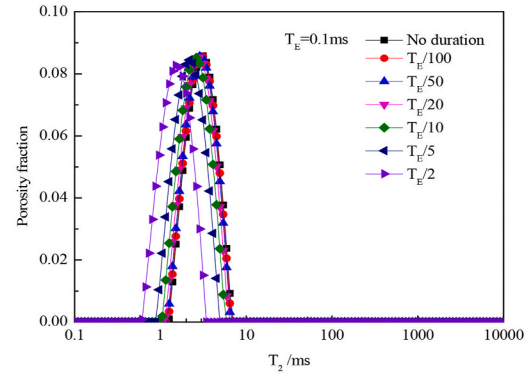
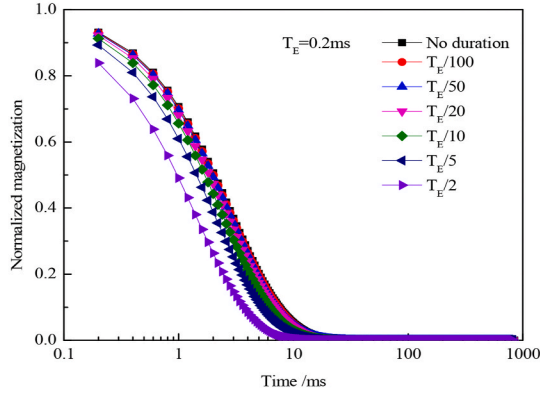
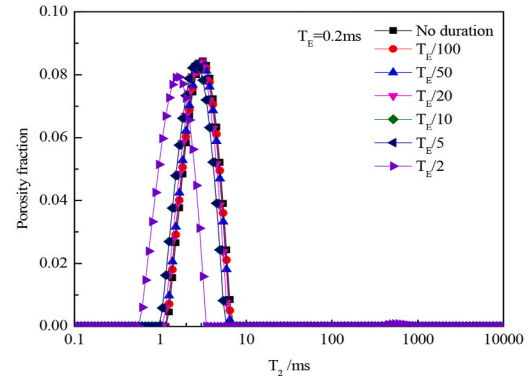
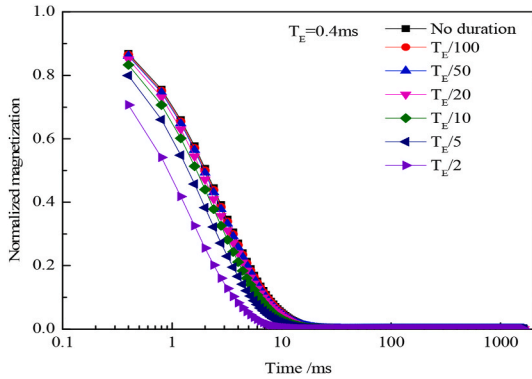
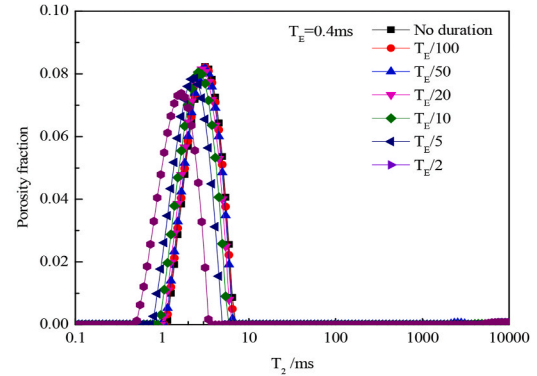
(a) Decay curves of $T_E = 0.1\text{ms}$ (b) T_2 distributions of $T_E = 0.1\text{ms}$ (c) Decay curves of $T_E = 0.2\text{ms}$ (d) T_2 distributions of $T_E = 0.2\text{ms}$ (e) Decay curves of $T_E = 0.4\text{ms}$ (f) T_2 distributions of $T_E = 0.4\text{ms}$

Fig. 10. Decay curves and inversion results of different pulse durations for the micropore.

non-uniformity of static magnet field due to hardware imperfections, chemical shifts, or effects of the radio frequency pulse sequence (Topgaard et al., 2005; Cai et al., 2009; Casabianca et al., 2014), and the internal magnetic field contributed by the magnetic susceptibility difference between the pore fluid and the rock matrix (Hurlimann, 1998; Winkler et al., 2003). For simplicity, the off-resonance effect introduced by the field inhomogeneity is not discussed here.

3. Numerical results

We discuss the NMR relaxation and the inversion result for two typical pores at different pulse properties. The forward T_2 spectrums are generated by the Gaussian function. To reduce the effect of the

longitudinal relaxation, the T_1 is fixed as three times as the T_2 , and the echo spacing is fixed as 0.1 ms. T_2 is logarithmically distributed from 0.1 ms to 1000 ms with 128 points. The peak values of both pores are 3 ms and 60 ms, respectively. Their T_2 distributions and NMR decay curves are shown in Fig. 4.

In the following paragraphs, we will discuss the influence of the excitation angle, the refocusing angle, the phase angle, as well as the pulse duration on the NMR signals. The diagram of the simulating parameters is shown in Fig. 5. For simplicity, we only give the NMR decay curve of the y-axis since the primary purpose is to simulate the CPMG pulse sequence as $90_x-180_y-180_y-180_y$.

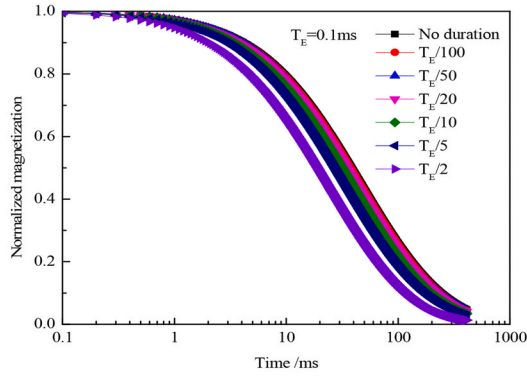
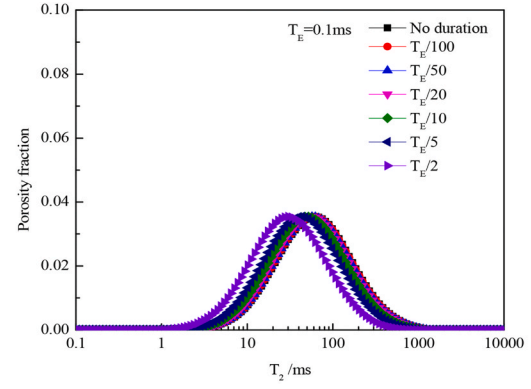
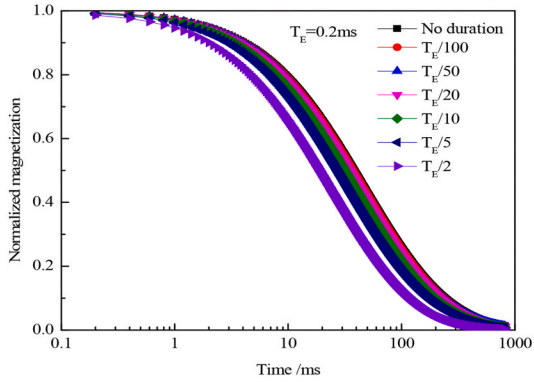
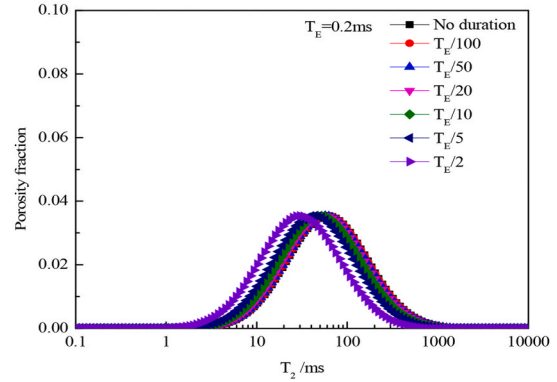
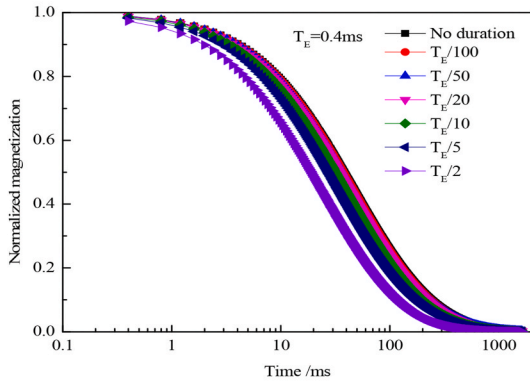
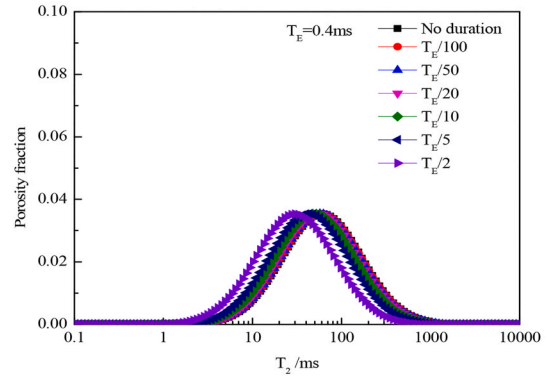
(a) Decay curves of $T_E = 0.1\text{ms}$ (b) T_2 distributions of $T_E = 0.1\text{ms}$ (c) Decay curves of $T_E = 0.2\text{ms}$ (d) T_2 distributions of $T_E = 0.2\text{ms}$ (e) Decay curves of $T_E = 0.4\text{ms}$ (f) T_2 distributions of $T_E = 0.4\text{ms}$

Fig. 11. Decay curves and inversion results of different pulse durations for the mesopore.

3.1. Effect of the excitation angle

Fig. 6a and Fig. 6b shows the relationship between the normalized magnetization of the first two echoes and the excitation angle. The normalized magnetization is symmetrical with the excitation angle (the value of θ is the same as $180-\theta$). A positive correlation between the normalized magnetization and the excitation angle is observed when the excitation angle is lower than 90° . Fig. 6c–f shows the simulated NMR decay curves and their inversion results at different excitation angles. It is clear that the amplitude is positively correlated with the excitation angle, revealing that the influence of the excitation angle should not be neglected. However, the magnetization variation caused by the excitation angle has weak influence on the T_2 distribution. It is also noted that

the amplitude of the T_2 spectrum simulated at the excitation angle of 90° is less than the forward micropore model, which is contributed by the decay during T_E .

3.2. Effect of the refocusing angle

Fig. 7 depicts the simulated NMR decay curves and their inversion results with different refocusing angles. It is obvious that the refocusing angle does not contribute any influence to the NMR response. It is easy to interpret since the main function of the refocusing pulse is to rephase the protons and generate the measurable signal. It is applicable to reduce the refocusing angle as long as the phase angle is stable. For simplicity, the inversion results are not shown.

3.3. Effect of the phase angle

Fig. 8 gives the simulated NMR decay curves and their inversion results for two pores using the standard CPMG pulse sequence but with different phase angles of the excitation pulse. It seen that the amplitude of the first and the second echo is negatively correlated with the phase angle. The decay curve and the corresponding T_2 spectrum alters slightly when the phase angle is lower than 15° , indicating that the influences of the phase angle can be neglected, both for the micropore and the mesopore. Fig. 9 shows the simulated NMR decay curves for two pores using the standard CPMG pulse sequence but with different phase angles of the refocusing pulse. The zig-zag phenomenon is observed for both pores when varying the phase angle. It is easy to explain that the time evolution of the magnetization vector is controlled by the phase angle and the duration of the refocusing pulse. Moreover, we found that the signal of the even echoes is not affected by the phase of the refocusing pulse. Therefore, choosing the data of even echoes may overcome the zig-zag phenomenon (or the tremble) brought by the phase angle of the refocusing pulse.

3.4. Effect of the pulse duration

Fig. 10 depicts the simulated NMR decay curves and theirs corresponding inversion results for the micropore using the standard CPMG pulse sequence but with different pulse durations. It is shown that the variation of the NMR response is slight at low pulse duration. However, the T_2 spectrum deviates from the forward model heavily when the pulse duration is larger than 20 percent of the echo spacing. Similar phenomenon is also observed for the mesopore, as is indicated in Fig. 11. Moreover, the mesopore is more sensitive to the pulse duration.

4. Conclusion

We developed a numerical algorithm to simulate the NMR responses of two representative pores based on the Bloch equation. The effects of excitation angles, refocusing angles, phase angles, pulse durations are investigated. Numerical tests on synthetic data show that the excitation angle is the controlling factor of NMR responses, but the refocusing angle is irrespective of the NMR response. The phase angle of the excitation pulse is negative with the amplitude and poses slight influence at low values. Moreover, the phase angle of the refocusing pulse only affects the amplitude of odd echoes, making it feasible to use the even echoes to reduce the inaccuracy. The effect of the pulse duration can be neglected with relative low values. However, its influence should be considered when the pulse duration is higher than 10% of the echo spacing, especially for the mesopore. However, much work should be done to incorporate the numerical simulation and the experimental result to verify our research. In additional, the effect of the off resonance and the inhomogeneity of the magnetic field are also to be investigated in future studies.

Data & code availability

The numerical code is available under the CC BY NC 3.0 license (free to adapt, copy or redistribute, providing you attribute appropriately and do not use the material for commercial use). The code was written in Matlab. The code used for this paper can be accessed at <https://github.com/gexinmin2002/NMR-simulation/>.

Authorship statement

Xinmin Ge led the research algorithm design, coding and writing. Yiren Fan assisted in writing and algorithm design. Jianyu Liu assisted the data analysis, coding, and writing. Jier Zhao assisted the coding and labelling. Binding Zeng assisted the data analysis.

Donghui Xing assisted the data analysis and pulse sequence.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Anand, V., 2017. Novel methodology for accurate resolution of fluid signatures from multi-dimensional NMR well-logging measurements. *J. Magn. Reson.* 276, 60–68.
- Andrade, F.D., Antonio, A.M., Colnago, L.A., 2011. Qualitative analysis by online nuclear magnetic resonance using Carr-Purcell-Meiboom-Gill sequence with low refocusing flip angles. *Talanta* 84, 84–88.
- Balac, S., Chupin, L., 2008. Fast approximate solution of Bloch equation for simulation of RF artifacts in magnetic resonance imaging. *Math. Comput. Model.* 48 (11), 1901–1913.
- Behroozmand, A.A., Keating, K., Auken, E., 2015. A review of the principles and applications of the NMR technique for near-surface characterization. *Surv. Geophys.* 36 (1), 27–85.
- Berman, P., Levi, O., Parmet, Y., Saunders, M., Wiesman, Z., 2013. Laplace inversion of low-resolution NMR relaxometry data using sparse representation methods. *Concepts Magn. Reson.* 42 (3), 72–88.
- Bhalekar, S., Daftardar-Gejji, V., Baleanu, D., Magin, R., 2011. Fractional Bloch equation with delay. *Comput. Math. Appl.* 61, 1355–1365.
- Bitoun, J., Taquin, J., Sauzade, M., 1984. A computer algorithm for the simulation of any Nuclear Magnetic Resonance (NMR) imaging method. *Magn. Reson. Imag.* 2, 113–120.
- Borneman, T.W., Hürlimann, M.D., Cory, D.G., 2010. Application of optimal control to CPMG refocusing pulse design. *J. Magn. Reson.* 207 (2), 220–233.
- Butler, J.P., Reeds, J.A., Dawson, S.V., 1981. Estimating solutions of first kind integral equations with nonnegative constraints and optimal smoothing. *SIAM J. Numer. Anal.* 18, 381–397.
- Cai, J., Chen, Q., 2016. De-noising for NMR oil well logging signals based on empirical mode decomposition and independent component analysis. *Arab. J. Geosci.* 9 (1), 1–11.
- Cai, S., Zhang, W., Chen, Z., 2009. High-resolution solution NMR spectra in inhomogeneous magnetic fields. *Curr. Anal. Chem.* 5, 70–83.
- Casabianca, L.B., Mohr, D., Mandal, S., Song, Y.-Q., Frydman, L., 2014. Chirped CPMG for well-logging NMR applications. *J. Magn. Reson.* 242, 197–202.
- Coates, G.R., Xiao, L., Prammer, M.G., 1999. *NMR Logging: Principles and Applications*. Houston, Texas.
- Davies, M., Puy, G., Vandergheynst, P., Wiaux, Y., 2014. A compressed sensing framework for magnetic resonance fingerprinting. *SIAM J. Imag. Sci.* 7 (4), 2623–2656.
- Dunn, K.-J., Bergman, D.J., Latorraca, G.A., 2002. *Nuclear Magnetic Resonance: Petrophysical and Logging Applications*, first ed. Amsterdam, New York.
- Feng, C., Fu, J., Shi, Y., Li, G., Mao, Z., 2016. Predicting reservoir wettability via well logs. *J. Geophys. Eng.* 13, 234–241.
- Freedman, R., Heaton, N., 2004. Fluid characterization using nuclear magnetic resonance logging. *Petrophysics* 45 (3), 241–250.
- Freedman, R., Johnston, M., Morriss, C.E., Straley, C., Tutunjian, P.N., Vinegar, H.J., 1997. Hydrocarbon saturation and viscosity estimation from nmr logging in the belridge diatomite. *Log. Anal.* 38, 44–72.
- Gai, N.D., Stehning, C., Nacif, M., Blumke, D.A., 2013. Modified look-locker T_1 evaluation using Bloch simulations: human and phantom validation. *Magn. Reson. Med.* 69 (2), 329–336.
- Ge, X., Fan, Y., Zhu, X., Chen, Y., Li, R., 2015. Determination of nuclear magnetic resonance T_2 cutoff value based on multifractal theory—An application in sandstone with complex pore structure. *Geophysics* 80 (1), D11–D21.
- Ge, X., Liu, J., Xing, D., Deng, S., Cai, J., 2018. Laboratory investigation into the formation and dissociation process of gas hydrate by low-field NMR technique. *J. Geophys. Res. Sol. Earth* 123 (5), 3339–3346.
- Grunewald, E., Knight, R., 2011. The effect of pore size and magnetic susceptibility on the surface NMR relaxation parameter T_2^* . *Near Surf. Geophys.* 9 (2), 169–178.
- Guo, J., Xie, R., Zou, Y., Jin, G., Gao, L., Xu, C., 2018. A new method for NMR data inversion based on double-parameter regularization. *Geophysics* 83 (5), JM39–JM49.

- Hu, F., Zhou, C., Li, C., Xu, H., Zhou, F., Si, Z., 2016. Water spectrum method of NMR logging for identifying fluids. *Petrol. Explor. Dev.* 43 (2), 268–276.
- Hurlimann, M.D., 1998. Effective gradients in porous media due to susceptibility differences. *J. Magn. Reson.* 131, 232–240.
- Hurlimann, M.D., Griffin, D.D., 2000. Spin dynamics of Carr-Purcell-Meiboom-Gill-like sequences in grossly inhomogeneous B_0 and B_1 fields and application to NMR well logging. *J. Magn. Reson.* 143, 120–135.
- Jiang, H., Daigle, H., Tian, X., Pyrcz, M.J., Griffith, C., Zhang, B., 2019. A comparison of clustering algorithms applied to fluid characterization using NMR T_1 - T_2 maps of shale. *Comput. Geosci.* 126, 52–61.
- Jurczuk, K., Kretowski, M., Bellanger, J.-J., Eliat, P.-A., Saint-Jalmes, H., Bézy-Wendling, J., 2013. Computational modeling of MR flow imaging by the lattice Boltzmann method and Bloch equation. *Magn. Reson. Imag.* 31 (7), 1163–1173.
- Konuma, T., Nagadoi, A., Kurita, J., Ikegami, T., 2018. Analysis of artifacts caused by pulse imperfections in CPMG pulse trains in NMR relaxation dispersion experiments. *Magnetochem.* 4, 33.
- Koroleva, V.D.M., Mandal, S., Song, Y.-Q., Hurlimann, M.D., 2013. Broadband CPMG sequence with short composite refocusing pulses. *J. Magn. Reson.* 230, 64–75.
- Kose, R., Kose, K., 2017. GPU-optimized fast 3D MRI simulator for experimentally compatible pulse sequences. *J. Magn. Reson.* 281, 51–65.
- Li, X., Xiao, L.-Z., Liu, H.-B., Zhang, Z.-F., Guo, B.-X., Yu, H.-J., Zong, F.-R., 2013. Optimization of nuclear magnetic resonance refocusing pulses to enhance signal intensity in gradient B_0 field. *Acta Physica Sinica-Chinese Edition* 62 (14), 147602.
- Liang, C., Xiao, L., Zhou, C., Wang, H., Hu, F., Liao, G., Jia, Z., Liu, H., 2019. Wettability characterization of low-permeability reservoirs using nuclear magnetic resonance: an experimental study. *J. Petrol. Sci. Eng.* 178, 121–132.
- Mandal, S., Oh, S., Hurlimann, M.D., 2015. Absolute phase effects on CPMG-type pulse sequences. *J. Magn. Reson.* 261, 121–132.
- Marty, B., Vignaud, A., Greiser, A., Robert, B., Sousa, P.L.d., Carlier, P.G., 2015. Bloch equations-based reconstruction of myocardium T_1 maps from modified look-locker inversion recovery sequence. *PLoS One* 10 (5), e0126766.
- Mehana, M., El-monier, I., 2016. Shale characteristics impact on. *Nucl. Magn. Res. (NMR) fluid typing methods and correlations: Petroleum* 2 (2), 138–147.
- Mitchell, J., Chandrasekera, T.C., 2014. Understanding generalized inversions of nuclear magnetic resonance transverse relaxation time in porous media. *J. Chem. Phys.* 141, 224201.
- Mitchell, J., Howe, A.M., Clarke, A., 2015. Real-time oil-saturation monitoring in rock cores with low-field NMR. *J. Magn. Reson.* 256, 34–42.
- Obasi, C., Pashin, J., 2018. Effects of internal gradients on pore-size distribution in shale. *AAPG (Am. Assoc. Pet. Geol.) Bull.* 102 (9), 1825–1840.
- Singh, H., 2016. A new numerical algorithm for fractional model of Bloch equation in nuclear magnetic resonance. *Alexandria Engineering Journal* 55 (3), 2863–2869.
- Song, Y.-Q., 2010. Recent progress of nuclear magnetic resonance applications in sandstones and carbonate rocks. *Vadose Zone J.* 9, 828–834.
- Tan, M., Fan, L., Mao, K., Li, J., Wu, C., 2019. Influential factors analysis and porosity correction method of nuclear magnetic resonance measurement in igneous rocks. *J. Appl. Geophys.* 161, 153–166.
- Testamanti, M.N., Rezaee, R., 2019. Considerations for the acquisition and inversion of NMR T_2 data in shales. *J. Petrol. Sci. Eng.* 174, 177–188.
- Topgaard, D., Sakellariou, D., Pines, A., 2005. NMR spectroscopy in inhomogeneous B_0 and B_1 fields with non-linear correlation. *J. Magn. Reson.* 175, 1–10.
- Walbrecker, J.O., Hertrich, M., Green, A.G., 2011. Off-resonance effects in surface nuclear magnetic resonance. *Geophysics* 76 (2), G1–G12.
- Wang, P., Venkataramanan, L., Jain, V., 2017. Sparse clustered bayesian-inspired T_1 - T_2 inversion from borehole NMR measurements. *IEEE Trans. Comput. Imag.* 3, 355–368.
- Washburn, K.E., 2014. Relaxation mechanisms and shales: concepts in. *Magnetic Resonance: Part. Accel.* 43A (3), 57–78.
- Winkler, M., Zhou, M., Bernardo, M., Endeward, B., Thomann, H., 2003. Internal magnetic gradient fields in glass bead packs from numerical simulations and constant time diffusion spin echo measurements. *Magn. Reson. Imag.* 21, 311–315.
- Xiao, L., Mao, Z., Wang, Z., Jin, Y., 2012. Application of NMR logs in tight gas reservoirs for formation evaluation: a case study of Sichuan basin in China. *J. Petrol. Sci. Eng.* 81, 182–195.
- Xie, R., Xiao, L., 2011. Advanced fluid-typing methods for NMR logging. *Petrol. Sci.* 8, 163–169.
- Xie, R., Wu, Y., Liu, K., Liu, M., Xiao, L., 2014. De-noising methods for NMR logging echo signals based on wavelet transform. *J. Geophys. Eng.* 11 (3), 035003.
- Yan, J.-P., He, X., Geng, B., Hu, Q.-H., Feng, C.-Z., Kou, X.-P., Li, X.-W., 2017. Nuclear magnetic resonance T_2 spectrum: multifractal characteristics and pore structure evaluation. *Appl. Geophys.* 14, 205–215.
- Zou, Y., Xie, R., 2015. A novel method for NMR data compression. *Comput. Geosci.* 19, 389–401.